application. Let's look at how we can do this tracking using GA.

```
…
fbLikeParams = {
    socialNetwork:
    'Facebook',
    socialAction:
    'Like',
    socialTarget:
    'https://facebook.com/mypage'
};
…
fbLike() {
    ga('send', 'social', this.
fbLikeParams);
}
```

If that code looks familiar, it's because it is very similar to the method by which we track button clicks. Let's look at the steps:

1. Construct the payload for sending data. This payload should have the following fields:
   a. *socialNetwork* – The network the interaction is happening with, e.g., Facebook, Twitter, etc.
   b. *socialAction* - What sort of interaction is happening, e.g., Like, Tweet, Share, etc.
   c. *socialTarget* - What URL is being targeted by using the interaction. This could be the URL of the social media profile/page.
2. The next method is, of course, to add a function to report this activity. Unlike a button click, we don't use the *send* method here, but the *social* method. Also, after this function is written, we bind it to the Like button we have in place.

There's more that can be done with GA as far as tracking user interactions go, one of the top items being exception tracking. This allows us to track errors and exceptions occurring in the application using GA. We won't delve deeper into it in this article; however, the reader is encouraged to explore this.

User identity

Privacy is a right, not a luxury: While Google Analytics can record a lot of activities as well as user interactions, there is one comparatively less known aspect, which we will look at in this section. There's a lot of control we can place over tracking (and not tracking) user identity.

Cookies: GA uses cookies as a means to track a user's activity. But we can define what these cookies are named and a couple of other aspects about them, as shown in the code snippet below:

```
trackingID =
    'UA-139883813-1'
    ;
  cookieParams = {
    cookieName: 'myGACookie',
    cookieDomain: window.location.
hostname,
    cookieExpires: 604800
  };
…
ngOnInit() {
     ga('create', this.trackingID,
this.cookieParams);
     ...
}
```

Here, we are setting the GA cookie's name, domain and cookie expiration date, which allows us to distinguish cookies set by our GA tracker from those set by GA trackers from other websites/ Web applications. Rather than a cryptic auto-generated identity, we'll be able to set custom identities for our application's GA tracker cookies.

IP anonymisation: There may be cases when we do not want to know where the traffic to our application is coming from. For instance, consider a button click activity tracker – we do not necessarily need to know the geographical source of that interaction, so long as the number of hits is tracked. In such situations, GA allows us to track users' activity without them having to reveal their IP address.

```
ipParams = {
    anonymizeIp: true
};
…
ngOnInit() {
     …
     ga('set', this.ipParams);
     ...
}
```

Here, we are setting the parameters of the GA tracker so that IP anonymisation is set to *true*. Thus, our user's IP address will not be tracked by Google Analytics, which will give users a sense of privacy.

Opt-out: At times, users may not want their browsing data to be tracked. GA allows for this too, and hence has the option to enable users to completely opt out of GA tracking.

```
...
optOut() {
    window['ga-disable-UA-139883813-1']
= true;
 }
...
```

optOut() is a custom function which disables GA tracking from the window. We can employ this function using event binding on a button/check box, which allows users to opt out of GA tracking.

We have looked at what makes integrating Google Analytics into single page applications tricky and explored a way to work around this. We also saw how we can track 'page' views in single page applications as well as touched upon tracking users' interactions with the application, such as button clicks, social media engagements, etc.

Finally, we examined opportunities offered by GA to ensure user privacy, especially when their identity isn't critical to our application's analytics, to the extent of allowing users to entirely opt out of Google Analytics tracking. Since there is much more that can be done, you're encouraged to keep exploring and keep playing around with the methods offered by GA. END

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